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Section : BAI-2A

Subject : MVC

Assignment : 4

Question 15 :

Sol :

$$= \int_1^4 \int_0^2 (6x^2y - 2x)\, dy \cdot dx$$

$$= \int_1^4 \left[3x^2y^2 - 2xy\right]_{y=0}^{y=2} \cdot dx$$

$$= \int_1^4 \left(3x^2(2)^2 - 2(2)x\right) - (0-0)\, dx$$

$$= \int_1^4 (12x^2 - 4x) \cdot dx$$

$$= \int \left(4x^3 - 2x\right)\Big]_{x=0}^{x=4} \cancel{dx}$$

$$= \left[4(4)^3 - 2(4)\right] - \cancel{4(1)^3 - 2(1)} \left[4(1)^3 - 2(1)\right]$$

$$= (256 - 32) - (4-2)$$

$$= 222$$

Question 17 :

Sol :

$$= \int_0^1 \int_1^2 (x + e^{-y}) \cdot dx \cdot dy$$

$$= \int_0^1 \left[\left(\frac{x^2}{2} + x \cdot e^{-y}\right)\right]_1^2 \cdot dy$$

$$= \int_0^1 \left[\frac{(2)^2}{2} + 2 \cdot e^{-y}\right] - \left[\frac{1^2}{2} + 1 \cdot e^{-y}\right] \cdot dy$$

$$= \int_0^1 (2 + 2e^{-y}) - \left(\frac{1}{2} + e^{-y}\right) \cdot dy$$

$$= \int_0^1 \left(\frac{3}{2} + e^{-y}\right) \cdot dy$$

$$= \left[\frac{3}{2}y - e^{-y}\right]_0^1$$

$$= \left[\frac{3}{2}(1) - e^{-1}\right] - \left[\frac{3}{2}(0) - e^{-(0)}\right]$$

$$= \left(\frac{3}{2} - e^{-1}\right) - (0-1).$$

$$= \frac{5}{2} - e^{-1}.$$

Question # 19

Sol:

$$= \int_{-3}^{3}\int_{0}^{\pi/2} (y + y^2\cos x).dx.dy$$

$$= \int_{-3}^{3}\left[xy + y^2.\sin x\right]_0^{\pi/2} .dy$$

$$= \int_{-3}^{3}\left[\frac{\pi}{2}(y) + y^2.\sin\left(\frac{\pi}{2}\right)\right] - \left[0(y) + y^2\sin(0)\right].dy$$

$$= \int_{-3}^{3}\left[\frac{\pi}{2}y + y^2(1)\right] - [0-0] .dy$$

$$= \int_{-3}^{3} \frac{\pi}{2}y + y^2 .dy$$

$$= \left[\frac{\pi}{4}y^2 + \frac{y^3}{3}\right]_{-3}^{3}$$

$$= \left[\frac{\pi}{4}(3)^2 + \frac{(3)^2}{3}\right] - \left[\frac{\pi}{4}(-3)^2 + \frac{(-3)^2}{3}\right]$$

$$= \left(\frac{9\pi}{4} + 9\right) - \left(\frac{9\pi}{4} - 9\right) \Rightarrow 18$$

Question # 23 :

Sol:

$$= \int_0^3 \int_0^{\pi/2} t^2 \cdot \sin^3\phi \cdot d\phi \, dt$$

$$= \int_0^3 t^2 \cdot dt \cdot \int_0^{\pi/2} \sin^3\phi \cdot d\phi \cdot$$

$$= \left[\frac{t^3}{3}\right]_0^3 \cdot \int_0^{\pi/2} \left(1 - \cos^2 \cdot \phi\right) \cdot \sin\phi \cdot d\phi$$

$$= \left[\frac{t^3}{3}\right]_0^3 \cdot \left[\frac{1}{3}\cos^3\phi - \cos\phi\right]_0^{\pi/2}$$

$$= \left[\frac{(3)^3}{3}\right] \cdot \left[\frac{1}{3}\cos^3\left(\frac{\pi}{2}\right) - \cos\left(\frac{\pi}{2}\right)\right] - \left[\frac{1}{3}\cos^3(0) - \cos(0)\right]$$

$$= \frac{\cancel{27}^{9}}{\cancel{3}_{1}} \cdot \left[\frac{1}{3}(0) - 0\right] - \left[\frac{1}{3} - 1\right]$$

$$= 9 \cdot \left(\frac{1}{3} - 1\right)$$

$$= \frac{2}{3}(9)$$

$$= 6$$

Question # 25

Sol:

$$= \int_0^1 \int_0^1 v\,(u + v^2)^4 \cdot du \cdot dv$$

$$= \int_0^1 \left[\frac{1}{5} v \cdot (u + v^2)^5\right]_{u=0}^{u=1} \cdot dv$$

$$= \frac{1}{5}\int_0^1 v \left[(1 + v^2)^5 - (0 + v^2)^5\right] \cdot dv$$

$$= \frac{1}{5}\int_0^1 \left[v(1+v^2)^5 - v^{11} \right] \cdot dv$$

$$= \frac{1}{5}\left[\frac{1}{2}v^2 \cdot \frac{1}{6}(1+v^2)^6 - \frac{1}{12}v^{12} \right]_0^1 \Rightarrow \frac{1}{5}\left[\frac{v^2(1+v^2)^6 - v^{12}}{12} \right]_0^1$$

Let $t = 1 + v^2$

$$\frac{dt}{dv} = 2v.$$

$$dt = 2v \cdot dv.$$

$$= \frac{1}{60}\left[(1+1)^6 - \frac{1}{1}(1)^{12} \right] - \left[(1+0)^6 - \frac{1}{1}(0)^{12} \right]$$

$$= \frac{1}{60}\left[(2^6 - 1) - (1 - 0) \right]$$

$$= \frac{1}{60}(63 - 1).$$

$$= \frac{31}{30}$$

Question 27:

Sol:

$$\iint_R x \cdot \sec^2 y \cdot dA, \quad R = \{(x,y) \mid 0 \le x \le 2,\ 0 \le y \le \pi/4\}$$

$$= \int_0^2 \int_0^{\pi/2} x \cdot \sec^2 y \cdot dy \cdot dx$$

$$= \int_0^2 x \cdot dx \cdot \int_0^{\pi/2} \sec^2 y \cdot dy.$$

$$= \left[\frac{x^2}{2} \right]_0^2 \cdot \left[\tan y \right]_0^{\pi/4}$$

$$= (2-0)\cdot\left(\tan\frac{\pi}{4} - \tan 0\right).$$

$$= 2(1-0) \Rightarrow 2$$

Question # 29

Sol:

$$\iint_R \frac{xy^2}{x^2+1}\,dA \,,\quad R = \{(x,y) \mid 0 \le x \le 1 ,\ -3 \le y \le 3\}$$

$$= \int_0^1 \int_3^{-3} \frac{xy^2}{x^2+1}\,dy\cdot dx$$

$$= \int_0^1 \frac{x}{x^2+1}\,dx \cdot \int_3^{-3} y^2\,dy$$

$$= \left[\frac{1}{2}\ln(x^2+1)\right]_0^1 \cdot \left[\frac{1}{3}y^3\right]_3^{-3}$$

$$= \frac{1}{2}\left[\ln(2) - \ln(1)\right] - \frac{1}{3}(27+27)$$

$$= 9\ln(2).$$

Question # 33:

Sol:

$$\iint_R y\cdot e^{-xy}\,dA \,,\quad R = [0,2] \times [0,3]$$

$$= \int_0^3 \int_0^2 y\cdot e^{-xy}\,dx\cdot dy.$$

$$= \int_0^3 \left[e^{-xy}\right]_0^2 dy$$

$$= \int_0^3 \left(e^{-2y} + 1\right)\cdot dy$$

$$= \left[\frac{1}{2} - e^{-2y} + y\right]_0^3$$

$$= \frac{1}{2}e^{-6} + 3 - \left(\frac{1}{2} + 0\right)$$

$$= \frac{1}{2}e^{-6} + \frac{5}{2}$$

Question # 34

Sol:

$$\iint_R \frac{1}{1+x+y}\, dA, \quad R = [1,3] \times [1,2]$$

$$= \int_1^3 \int_1^2 \frac{1}{1+x+y} \cdot dy \cdot dx$$

$$= \int_1^3 \left[\ln(1+x+y)\right]_1^2$$

$$= \int_1^3 [\ln(x+3) - \ln(x+2)] \cdot dx$$

$$= \int_1^3 \left((x+3)\ln(x+3) - (x+3)\right) - \left((x+2)\ln(x+2) - (x+2)\right]_1^3$$

Formula : $\int \ln(a+y) \cdot dy = (a+y)\ln(a+y) \cdot - (a-y)$

$$= \left[(x+3)\ln(x+3) - (x+3)\right]_1^3 - \left[(x+2)\ln(x+2) - (x+2)\right]_1^3$$

$$= \left\{\left[(3+3)\ln(3+3) - (3+3)\right] - \left[(3+2)\ln(3+2) - (3+2)\right]\right\}$$

$$- \left\{\left[(1+3)\ln(1+3) - (1+3)\right] - \left[(1+2)\ln(1+2) - (1+2)\right]\right\}$$

$$= (6\ln 6 - 6 - 5\ln(5) + 5) - (4\ln(4) - 4 - 3\ln(3) + 3)$$

$$= 6\ln 6 - 5\ln 5 - 4\ln 4 + 3\ln(3).$$

$$=$$

Question 43 :

volume = ? lies under the
plane → $4x + 6y - 2z + 15 = 0$
&
lies above the rectangle
$R = \{ (x,y) \mid -1 \leq x \leq 2 , \; -1 \leq y \leq 1 \}$

plane :

$$4x + 6y - 2z + 15 = 0$$

or

$$z = 2x + 3y + \frac{15}{2}$$

Volume = ?

$$V = \iint_R \left(2x + 3y + \frac{15}{2} \right) dA$$

$$V = \int_{-1}^{+1} \int_{-1}^{2} \left(2x + 3y + \frac{15}{2} \right) \cdot dx \cdot dy$$

$$V = \int_{-1}^{+1} \left[\left(x^2 + 3xy + \frac{15}{2} x \right) \right]_{-1}^{2} dy$$

$$V = \int_{-1}^{+1} \left[2^2 + 3(2)y + \frac{15}{2}(2) \right] - \left[(-1)^2 + 3(-1)y + \frac{15(-1)}{2} \right] \cdot dy$$

$$V = \int_{-1}^{+1} \left(4 + 6y + 15 - 1 + 3y + \frac{15}{2} \right) dy$$

$$V = \int_{-1}^{+1} (19 + 6y) - \frac{13}{2} + 3y \cdot dy$$

$$V = \int_{-1}^{+1} \left(9y + \frac{51}{2} \right) dy$$

$$V = \int_{-1}^{1} \left[\frac{51}{2}y + \frac{9}{2}y^2 \right]_{-1}^{1}$$

$$V = \left[\frac{51}{2}(1) + \frac{9}{2}(1)^2\right] - \left[\frac{51}{2}(-1) + \frac{9}{2}(-1)^2\right]$$

$$= \left(\frac{51}{2} + \frac{9}{2}\right) - \left(-\frac{51}{2} + \frac{9}{2}\right)$$

$$= \left(\frac{60}{2}\right) - \left(-\frac{42}{2}\right)$$

$$= 30 - (-21)$$

$$= 51$$

Question 47 :

Sol:

volume of solid enclosed by the surface

$z = 1 + x^2 y \cdot e^y$

and planes $z = 0$, $x = \pm 1$, $y = 0$

and $y = 1$

$$V = \iint_R \left(1 + x^2 y \cdot e^y\right) \cdot dA$$

$$V = \int_0^1 \int_{-1}^{1} (1 + x^2 y e^y)\, dx \cdot dy$$

$$= \int_0^1 \left[\left(x + \frac{1}{3}x^3 y \cdot e^y\right)\right]_{x=-1}^{x=1} \cdot dy$$

$$= \int_0^1 \left(1 + \frac{1}{3}(1)^3 \cdot y \cdot e^y\right) - \left(-1 + \frac{1}{3}(-1)^3 \cdot y \cdot e^{+y}\right) \cdot dy$$

$$= \int 1 + \frac{1}{3}y \cdot e^y + 1 + \frac{1}{3}y \cdot e^{+y}$$

$$= \int_0^1 \left(2 + \frac{2}{3} y.e^y\right).dy \Rightarrow 2y + \frac{2}{3}\left[y.e^y - \int e^y.dy\right.$$

$$\Rightarrow 2y + \frac{2}{3}(ye^y - e^y).$$

$$= \left[2y + \frac{2}{3}(y-1).e^y\right]_0^1$$

$$= \left(2(1) + \frac{2}{3}(1-1).e^y\right) - \left(0 - \frac{2}{3}(-1)e^0\right)$$

$$= (2) - \left(-\frac{2}{3}e^0\right).$$

$$= 2 + \frac{2}{3} \Rightarrow \frac{8}{3}$$

Exercise 15.2 :

Question # 3

$$= \int_0^1\int_0^y x.e^{y^3}.dx.dy$$

$$= \int_0^1 \left[\frac{1}{2}x^2.e^{y^3}\right]_{x=0}^{x=y}.dy$$

$$= \int_0^1 \frac{1}{2}.e^{y^3}\left[(y)^2 - (0)^2\right].dy$$

$$= \int_0^1 \frac{1}{2}e^{y^3}.y^2.dy$$

$$= \quad \text{Let } u = y^3$$

$$= \quad \frac{du}{dy} = 3y^2$$

$$du = 3y^2.dy.$$

$$\frac{du}{3} = y^2.dy$$

$$\frac{du}{6} = \frac{1}{2}y^2.du.$$

So,

$$= \frac{1}{6}\int_0^1 e^u \cdot du$$

$$= \frac{1}{6}\left[e^u\right]_0^1$$

$$= \frac{1}{6}\left(e^1 - e^0\right)$$

$$= \frac{1}{6}(e-1).$$

Question # 5

Sol:

$$= \int_0^1\int_0^{s^2} \cos(s^3) \cdot dt \cdot ds.$$

$$= \int_0^1 \left[t \cdot \cos(s^3)\right]_{t=0}^{t=s^2} ds$$

$$= \int_0^1 s^2 \cdot \cos(s^3) \cdot ds$$

Let

$$u = s^3.$$

$$du = 3s^2 \cdot ds$$

$$\frac{du}{3} = s^2 \cdot ds.$$

$$= \int_0^1 \frac{\cos(u)}{3} \cdot du$$

$$= \frac{1}{3}\int_0^1 \cos(u) \cdot du$$

$$= \frac{1}{3}\left[\sin(u)\right]_0^1$$

$$= \frac{1}{3}\sin(1).$$

Question 9:

Sol:

$$f(x,y) = xy.$$

$y = \sqrt{x}$, $y = x - 2$

so, when curves interest then,

$$\sqrt{x} = x - 2$$

$$x = x^2 + 4 - 4x$$

$$x^2 - 5x + 4 = 0.$$

$$(x-4)(x-1) = 0.$$

then

$$x = 1 \quad , \quad x = 4.$$

The point for $x=1$ is not in D.

Thus, the point of intersection of curves is $(4,2)$ and the integral is

$$= \int_0^2 \int_{y^2}^{y+2} xy \cdot dx \cdot dy.$$

Question 11:

Sol:

$$\iint_D \frac{y}{x^2+1} \cdot dA \; , \quad D = \{(x,y) \mid 0 \le x \le 4 , 0 \le y \le \sqrt{x}\}$$

$$= \int_0^4 \int_0^{\sqrt{x}} \frac{y}{\cancel{y}^2+1} \cdot dy \cdot dx$$

$$= \int_0^4 \left[\frac{y^2}{2} \cdot \frac{1}{x^2+1}\right]_0^{\sqrt{x}} dx$$

$$= \frac{1}{2} \int_0^4 \frac{1}{x^2+1} \left[y^2\right]_0^{\sqrt{x}}$$

$$= \frac{1}{2}\int_0^4 \frac{1}{x^2+1}\cdot(\sqrt{x})^2\cdot dx$$

$$= \frac{1}{2}\int_0^4 \frac{x}{x^2+1}\cdot dx$$

$$= \frac{1}{2}\left[\ln(x^2+1)\cdot\frac{1}{2}\right]_0^4$$

$$= \frac{1}{4}\left[\ln(4^2+1)-\ln(0^2+1)\right]$$

$$= \frac{1}{4}\left(\ln(17)-\ln(1)\right).$$

$$= \frac{1}{4}\ln(17).$$

Question 13:

Sol:

$$\iint_D e^{-y^2}\cdot dA\,,\quad D=\{(x,y)\mid 0\le y\le 3,\ 0\le x\le y\}.$$

$$= \int_0^3\int_0^y e^{-y^2}\cdot dx\,dy.$$

$$= \int_0^3 \left[x\cdot e^{-y^2}\right]_0^y dy$$

$$= \int_0^3 y\cdot e^{-y^2}\cdot dy$$

$$= \left[\frac{-1}{2}e^{-y^2}\right]_0^3$$

$$= \frac{-1}{2}\left[e^{-3^2}-e^{-0}\right]$$

$$= \frac{-1}{2}\left(e^{-9}-1\right)$$

$$= \frac{1}{2}\left(1-e^{9}\right)$$

Question 19

Sol:

X

$\iint_D y \cdot dA$, D is bounded by $y = x - 2$, $x = y^2$

The curves $y = x^2$ and $y = 3x$ intersect at point $(0,0)$, $(3,9)$

$y = x^2$ and $y = 3x$.

so,

$x^2 = 3x$

$= 3x - x^2$.

$x(3-x) = 0$

$x = 0$ $\quad$ $3 = x$

or $x = 3$

Question 19 :

Sol:

$\iint_D y \cdot dA$, D is bounded by $y = x - 2$, $x = y^2$

$y = x - 2$, $x = y^2$

$y = y^2 - 2$

or

$y^2 = y + 2$

$y^2 - y - 2 = 0$

$(y-2)(y+1) = 0$

$y = -1$, $y = 2$

So, the point of intersection are $(1,-1)$ and $(4,2)$

Type I Region:

→ Upper borendry curve is $y = \sqrt{x}$

→ Lower boundry curve consists of two parts.

→ $y = -\sqrt{x}$ for $0 \leq x \leq 1$

→ $y = x - 2$ for $1 \leq x \leq 4$.

→ Domain = ?

$$D = \{(x,y) \mid 0 \leq x \leq 1,\ -\sqrt{x} \leq y \leq \sqrt{x}\} \cup \{(x,y) \mid 1 \leq x \leq 4,\ x-2 \leq y \leq \sqrt{x}\}$$

and

$$\iint_D y \cdot dA = \int_0^1 \int_{-\sqrt{x}}^{\sqrt{x}} y \cdot dy \cdot dx + \int_1^4 \int_{x-2}^{\sqrt{x}} y \cdot dy \cdot dx.$$

Type II region:

→ Left boundry $x = y^2$

→ Right boundry $x = y + 2$ for $-1 \leq y \leq 2$

→ Domain = ?

$$D = \{(x,y) \mid -1 \leq y \leq 2,\ y^2 \leq x \leq y+2\}$$

and

$$\iint_D y \cdot dA = \int_{-1}^{2} \int_{y^2}^{y+2} y \cdot dx \cdot dy$$

$$\iint_D y \cdot dA = \int_{-1}^{2}\int_{y}^{y+2} y \cdot dx \cdot dy$$

$$= \int_{-1}^{2} [xy]_{x=y^2}^{x=y+2} \cdot dy$$

$$= \int_{-1}^{2} (y+2)y - y^2(y) \cdot dy$$

$$= \int_{1}^{2} (y^2 + 2y - y^3) \cdot dy$$

$$= \left[\frac{y^3}{3} + y^2 - \frac{y^4}{4}\right]_{-1}^{2} \cancel{\cdot dy}$$

$$= \left(\frac{2^3}{3} + 2^2 - \frac{2^4}{4}\right) - \left(\frac{(-1)^3}{3} + (-1)^2 - \frac{(-1)^4}{4}\right)$$

$$= \left(\frac{8}{3} + \cancel{4} - \cancel{4}\right) - \left(\frac{-1}{3} + 1 - \frac{1}{4}\right)$$

$$= \frac{8}{3} - \left(\frac{-7}{12} + 1\right)$$

$$= \frac{8}{3} - \left(\frac{5}{12}\right)$$

$$= \frac{\cancel{27}^{9}}{\cancel{12}_{4}} \Rightarrow \frac{9}{4}$$

Question 23:

Sol:

$$\iint_D x \cdot \cos y \cdot dA \text{ . D is bounded by}$$

$$y = 0 \text{ , } y = x^2 \text{ , } x = 1$$

$$= \int_0^1 \int_0^{x^2} x \cdot \cos y \cdot dy \cdot dx$$

$$= \int_0^1 \left[x \cdot \sin y \right]_{y=0}^{y=x^2} \cdot dx$$

$$= \int_0^1 x \left[\sin x^2 - \sin 0 \right] \cdot dx$$

$$= \int_0^1 x \cdot \sin x^2 \cdot dx \quad \rightarrow \text{Let } u = x^2 \text{ . } du = 2x \cdot dx \quad x \cdot dx = \frac{1}{2} du$$

$$= -\frac{1}{2} \left[\cos(u) \right]_0^1 du$$

$$= -\frac{1}{2} \left[\cos(1) - \cos(0) \right]$$

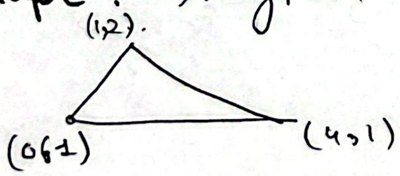
$$= +\frac{1}{2} (1 - \cos 1) .$$

Question 25:

Sol:

$$\iint_D y^2 \cdot dA \text{ , D is triangular}$$

Shape: region with vertices $(0,1)$, $(1,2)$, $(4,1)$



so, bottom : $y = 1$.

Top edges are two lines. So,

→ **Find equations:**

- **Left line:** $(0, 1) \rightarrow (1, 2)$

$$y = mx + c$$

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{2-1}{1-0} = 1$$

$(0,1) \rightarrow$ $1 = (1)0 + c \Rightarrow c = 1$

then

$$y = 1x + 1$$

$$y = x + 1 \quad \text{or} \quad x = y - 1$$

- **Right line:** $(1, 2) \rightarrow (4, 1)$

$$y = mx + c$$

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{1-2}{4-1} \Rightarrow \frac{-1}{3}$$

$(1,2) \rightarrow$

$$2 = \frac{-1}{3}(1) + c$$

$$c = \frac{7}{3}$$

$$y = \frac{-1}{3}x + \frac{7}{3} \quad \text{or} \quad x = 7 - 3y.$$

→ **Split region:**

because top changes at $x = 1$.

Limits:

- For left part:

$$0 \le x \le 1, \quad 1 \le y \le x + 1$$

- For right part:

$$1 \le x \le 4, \quad 1 \le y \le -\frac{x}{3} + \frac{7}{3}$$

Integrals :

$$= \int_0^1 \int_1^{x+1} y^2 \cdot dy \cdot dx + \int_1^4 \int_1^{-\frac{x}{3}+7/3} y^2 \cdot dy \cdot dx .$$

$$= \int_0^1 \left[\frac{y^3}{3} \right]_{y=1}^{y=x+1} \cdot dx + \int_1^4 \left[\frac{y^3}{3} \right]_{y=1}^{y=\frac{-x}{3}+\frac{7}{3}} \cdot dx$$

$$= \frac{1}{3} \int_0^1 \left[(x+1)^3 - (1)^3 \right] dx + \frac{1}{3} \int_1^4 \left[\left[\frac{-x}{3} + \frac{7}{3} \right]^3 - (1)^3 \right] \cdot dx$$

$$= \frac{1}{3} \int_0^1$$

Integrals :

$$\iint_D y^2 \cdot dA = \int_1^2 \int_{x=y-1}^{x=7-3y} y^2 \cdot dx \cdot dy$$

$$= \int_1^2 \left[\frac{y^3}{3} \right]_{x=y-1}^{x=7-3y} \cdot dy$$

$$= \frac{1}{3} \int_1^2 \left[y^3 \right]_{x=y-1}^{x=7-3y} \cdot dy$$

$$= \frac{1}{3} \int_1^2 (7-3y)^3 -$$

$$= \int_1^2 \left[xy^2 \right]_{x=y-1}^{x=7-3y} \cdot dy$$

$$= \int_1^2 \left[(7-3y) - (y-1) \right] y^2 \cdot dy$$

$$= \int_1^2 (7 - 3y - y + 1) y^2 \cdot dy$$

$$= \int_1^2 (-4y + 8) y^2 \cdot dy$$

$$= \int_1^2 (8y^2 - 4y^3) \cdot dy .$$

$$= \left[\frac{8y^3}{3} - y^4 \right]_1^2 \cdot \cancel{dy}$$

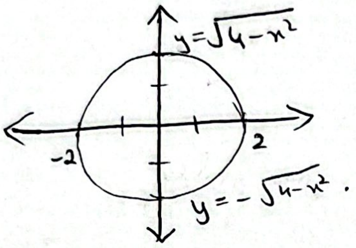
$$= \left[\left(\frac{8(2)^3}{3} - (2)^2 \right) - \left(\frac{8(1)^3}{3} - (1)^4 \right) \right]$$

$$= \frac{64}{3} - 16 - \frac{8}{3} + 1 \Rightarrow \frac{11}{3}.$$

Question 27:

Sol:

$\iint_D (2x - y)\cdot dA$, D is bounded by the circle with center the origin and radius 2.



$$x^2 + y^2 = (2)^2$$

$$x^2 + y^2 = 4.$$

$$y^2 = 4 - x^2$$

$$y = \sqrt{4 - x^2}.$$

$$= \int_{-2}^{2} \int_{y=\sqrt{4-x^2}}^{y=\sqrt{4-x^2}} (2x - y)\cdot dy \cdot dx$$

$$= \int_{-2}^{2} \left[2xy - \frac{y^2}{2} \right]_{\sqrt{4-x^2}}^{\sqrt{4-x^2}} \cdot dx$$

$$= \int_{-2}^{2} \left[2x\sqrt{4-x^2} - \frac{1}{2}(\cancel{4-x^2}) + 2x\sqrt{4-x^2} + \frac{1}{2}(\cancel{4-x^2}) \right] \cdot dx$$

$$= \int_{-2}^{2} 4x\sqrt{4-x^2} \cdot dx$$

$$= \frac{-4}{3} (4-x^2)^{3/2} \Big]_{-2}^{2} = 0$$

Note:

$4x\sqrt{4-x^2}$ is odd function

so $\int_{-2}^{2} 4x\sqrt{4-x^2} \cdot dx = 0.$

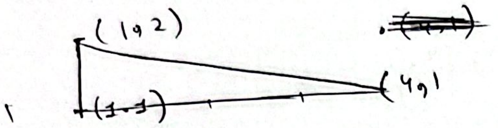
Question 33 :

→ Under the surface,

$z = xy$

→ above the triangle with vertices $(1,1), (4,1), (1,2)$

Volume = ?



Limits :

we know that

y = ? top → $y = 2$

bottom → $y = 1$

then $x = ?$

$(1,2) \longrightarrow (4,1)$

$$m = \frac{1-2}{4-1} \Rightarrow \frac{-1}{3}$$

$$2 = \frac{-1}{3}(1) + c \Rightarrow c = 7$$

Equation = ?

$$y = \frac{-1}{3}x + 7$$

$$y = \frac{-1}{3}x + 7$$

$$3y = -x$$

$$3y - 7 = -x$$

$$x = 7 - 3y.$$

$$\int_1^2 \int_1^{7-3y} x \cdot y \; dx \cdot dy$$

$$= \int_1^2 \left[\frac{x^2}{2} y \right]_{x=1}^{x=7-3y} \cdot dy$$

$$= \frac{1}{2} \int_1^2 \left[(7-3y)^2 - (1)^2 \right] y \cdot dy$$

$$= \frac{1}{2} \int_1^2 (48 + 9y^2 - 42y) y \, dy$$

$$= \frac{1}{2} \int_1^2 (48y + 9y^3 - 42y^2) \cdot dy$$

$$= \frac{1}{2} \left[\frac{48y^2}{2} + \frac{9y^4}{4} - \frac{42y^3}{3} \right]_1^2$$

$$= \frac{1}{2} \left[24y^2 - 14y^3 + \frac{9}{4} y^4 \right]_1^2$$

$$= \frac{1}{2} \left[\left(24(2)^2 - 14(2)^3 + \frac{9}{4}(2)^4 \right) - \left(24(1) - 14(1)^3 + \frac{9}{4}(1)^4 \right) \right]$$

$$= \frac{31}{8}$$

Exercise 15.3

Question 9:

$\iint_D x^2 y \cdot dA$, where D is the top half of the disk center the origin & radius 5.

Domain, D = ?

$\iint D = \{(r,\theta) \mid 0 \le r \le 5, \; 0 \le \theta \le \pi\}$.

Then,

$$\iint_D x^2 y \cdot dA = \int_0^{\pi}\int_0^5 (r\cos\theta)^2 (r\cdot\sin\theta)\cdot r\, dr\cdot d\theta$$

$$= \int_0^{\pi} \cos^2\theta \cdot \sin\theta \cdot d\theta \cdot \int_0^5 r^4 \cdot dr$$

$$= \left[\frac{-1}{3}\cos^3\theta\right]_0^{\pi} \cdot \left[\frac{r^5}{5}\right]_0^5$$

$$= \frac{-1}{3}\left[\cos^3(\pi) - \cos^3(0)\right] \frac{1}{5}\left[(5)^5 - (0)^5\right].$$

$$= \frac{-1}{3}(-1-1)\cdot 625$$

$$= \frac{1250}{3}.$$

Question 11:

Sol:

$\iint_R \sin(x^2+y^2)\cdot dA$, where R is in 1^{st} quadrant b/w circle ~~$x^2+y^2=4$~~ with center origin & radii 1 and 3.

$$= \int_0^{\pi/2} \int_1^{3} \sin(x^2+y^2) \cdot dA.$$

$$= \int_0^{\pi/2} \int_1^{3} \sin(r^2) \cdot r \cdot dr \cdot d\theta.$$

$$= \int_0^{\pi/2} d\theta \cdot \int_1^{3} \sin(r^2) \cdot r \cdot dr.$$

Let $r^2 = u$
$du = 2r \cdot dr$
$\frac{du}{2} = r \, dr$

$$= [\theta]_0^{\pi/2} \cdot \int_1^3 \frac{1}{2} \sin(u^2) \cdot du.$$

$$= [\theta]_0^{\pi/2} \cdot \frac{1}{2} \left[-\cos(u^2) \right]_1^3$$

$$= \left(\pi/2 - 0\right) - \frac{1}{2} \left[\cos(3^2) - \cos(1^3) \right]$$

$$= \left(\frac{\pi}{2}\right) \cdot \frac{-1}{2} (\cos 9 - \cos 1).$$

$$= \frac{\pi}{4} (\cos 1 - \cos 9).$$

Question 13:

$\iint_D e^{-x^2-y^2} \cdot dA$, where D is region bounded by the semicircle $x = \sqrt{4-y^2}$ and the y-axis

so

radius : $1 \to 2$.

$$= \int_{-\pi/2}^{\pi/2}\int_0^2 e^{-r^2} \cdot r \cdot dr \cdot d\theta$$

$$= \int_{-\pi/2}^{\pi/2} d\theta \cdot \int_0^2 e^{-r^2} \cdot r \cdot d\theta$$

Let

$$u = -r^2$$

$$du = -2r \cdot dr$$

$$-\frac{du}{2} = r \cdot dr$$

$$= [\theta]_{-\pi/2}^{\pi/2} \cdot \frac{-1}{2}\int_0^2 e^u \cdot du$$

$$= \left[\frac{\pi}{2} - \left(\frac{-\pi}{2}\right)\right] \cdot -\left[\frac{1}{2} e^u\right]_0^2$$

$$= \pi \cdot \left[\frac{-1}{2} \cdot e^{-r^2}\right]$$

$$= -\frac{\pi}{2} \cdot \left(e^{-2^2} - e^0\right)$$

$$= \frac{+\pi}{2} \cdot \left(1 - e^4\right).$$

Question 17:

Sol:

$$r = 1 - \cos\theta$$

$$D = \{(r, \theta) \mid 0 \leq r \leq 1 - \cos\theta,\ 0 \leq \theta \leq \pi/2\}$$

so, total area is

$$4A(D) = 4\int_0^{\pi/2}\int_0^{1-\cos\theta} r \cdot dr \cdot d\theta$$

$$= 4\int_0^{\pi/2}\left[\frac{r^2}{2}\right]_0^{1-\cos\theta}$$

$$= 2\int_0^{\pi/2} (1-\cos\theta)^2 d\theta$$

$$= 2\int_0^{\pi/2} (1 - 2\cos\theta + \cos^2\theta).d\theta.$$

$$= 2\int_0^{\pi/2} 1 - 2\cos\theta + \frac{1}{2}(1+\cos 2\theta).$$

$$\because \cos^2\theta = \frac{1+\cos 2\theta}{2}$$

$$= 2\left[\theta - 2\sin\theta + \frac{1}{2}\theta + \frac{1}{2}\left(\frac{\sin 2\theta}{2}\right)\right]_0^{\pi/2}$$

$$= 2\left[\theta - 2\sin\theta + \frac{1}{2}\theta + \frac{\sin 2\theta}{4}\right]_0^{\pi/2}$$

$$= 2\left(\frac{\pi}{2} - 2\sin\left(\frac{\pi}{2}\right) + \frac{1}{2}\left(\frac{\pi}{2}\right) + \frac{\sin 2(\pi/2)}{4}\right).$$

$$= 2\left(\frac{\pi}{2} - 2 + \frac{\pi}{4}\right) \Rightarrow \frac{3\pi}{2} - 4.$$

Question 39 :

Sol :

$$= \int_0^2 \int_0^{\sqrt{4-x^2}} . e^{-x^2-y^2}.dy\,dx$$

$y = \sqrt{4-x^2}$

$x^2 + y^2 = 4$

$$= \int_0^{\pi/2} \int_0^2 e^{-r^2}.dr.d\theta$$

Let $u = -r^2$

$du = -2r\,dr$

$-\frac{1}{2}du = r.dr.$

$$= \int_0^{\pi/2} d\theta . \int_0^2 re^{-r^2}.dr.$$

$$= [\theta]_0^{\pi/2} . -\frac{1}{2}\int_0^2 e^u .du$$

$$= \frac{\pi}{2}\left[-\frac{1}{2}e^{-r^2}\right]_0^2$$

$$= \frac{\pi}{4}(1 - e^{-4}).$$

Exercise 15.5 :

Question 5 :

Sol: part of plane $3x + 2y + z = 6$ lies in first octant.

$$z = 6 - 3x - 2y.$$

→ surface Area formula :

$$A(S) = \iint_D \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2} \cdot dA.$$

→ Finding limits :

In first octant, $z \geq 0$.

so, $3x + 2y \leq 6$.

Find intercepts :

If $y = 0$, $3x = 6 \Rightarrow x = 2$

If $x = 0$, $2y \leq 6 \Rightarrow y = 3$.

solve for y : convert boundry into limits

$$3x + 2y \leq 6$$

$$2y \leq 6 - 3x.$$

$$y \leq 3 - \frac{3x}{2}.$$

→ $f_x(x,y) = \frac{d}{dx}\, 6 - 3x - 2y$

$= -3$

→ $f_y(x,y) = \frac{d}{dx}\, 6 - 3x - 2y$

$= -2.$

$$A(S) = \iint_D \sqrt{(-3)^2+(-2)^2+1} \cdot dA$$

$$= \iint_D \sqrt{14} \cdot dA$$

$$= \sqrt{14} \iint_D dA.$$

$$= \sqrt{14} \left(\frac{1}{2} \cdot 2 \cdot 3\right)$$

$$= 3\sqrt{14}$$

Question 7:

Sol:

$z = 1 - x^2 - y^2$ ~~~~ where $z = -2$

$$-2 = 1 - x^2 - y^2$$

$$-3 = -x^2 - y^2$$

$$x^2 + y^2 = 3.$$

$$A(S) = \iint_D \sqrt{(f_x)^2 + (f_y)^2 + 1}$$

$$= \iint_D \sqrt{(-2x)^2 + (-2y)^2 + 1}$$

$$= \iint_D \sqrt{4(x^2+y^2)+1} \cdot dA$$

$$= \int_0^{2\pi}\int_0^{\sqrt{3}} \sqrt{4r^2+1} \cdot r \cdot dr \cdot d\theta.$$

$$= \int_0^{2\pi} \theta \cdot \int_0^{\sqrt{3}} \sqrt{4r^2+1} \cdot r \, dr.$$

$$= \cancel{[\theta]_0^{2\pi} \left[\frac{1}{12}\right]^0}$$

Let $u = 4r^2 + 1$

$du = 8r \cdot dr$

$\frac{du}{8} = r \cdot dr$

$$= \int_0^{2\pi} \theta \cdot \int_0^{\sqrt{3}} \frac{1}{2} \cdot (u)^{1/2} \cdot du$$

$$= [\theta]_0^{2\pi} \frac{1}{4\cancel{8}} \cdot \frac{\cancel{2}}{3} \left[(u)^{3/2}\right]_0^{\sqrt{3}}$$

$$= [\theta]_0^{2\pi} \cdot \frac{1}{12} \left[(u)^{3/2}\right]_0^{\sqrt{3}}$$

$$\cancel{= [2\pi - 0] \cdot \frac{1}{12} \left[(4(\sqrt{3})^2 + 1)^{3/2} - (4(0)+1)\right]}$$

$$\cancel{= 2\pi \cdot \frac{1}{3} (13)^{3/2} - (1)^{3/2}}$$

$$\cancel{= \frac{2\pi}{3} (13\sqrt{\ }}$$

$$= (2\pi - 0) \cdot \frac{1}{12} (13^{3/2} - 1)$$

$$= \frac{\pi}{6} (13\sqrt{13} - 1).$$

Ans.

Question # 8

Sol:

$$x^2 + z^2 = 4$$

$$z = \sqrt{4 - x^2} \quad (\text{since } z \geq 0)$$

so,

$$f_x(x,y) = -x(4-x^2)^{-1/2}.$$

$$f_y(x,y) = 0.$$

and,

Surface Area:

$$A(S) = \int_0^1 \int_0^1 \sqrt{[-x(4-x^2)^{-1/2}]^2 + 0^2 + 1} \cdot dy\,dx.$$

$$A(S) = \int_0^1\int_0^1 \sqrt{\frac{+x}{4-x^2} + 1} \; dy \cdot dx$$

$$A(S) = \int_0^1\int_0^1 \sqrt{\frac{\cancel{x^2} + 4 - \cancel{x^2}}{4-x^2}} \cdot dy \cdot dx$$

$$= \int_0^1\int_0^1 \sqrt{\frac{4}{4-x^2}} \cdot dy \cdot dx.$$

$$= \int_0^1\int_0^1 \frac{2}{\sqrt{4-x^2}} \cdot dy \cdot dx$$

$$= \int_0^1 \cdot dy \cdot \int_0^1 \frac{2}{\sqrt{4-x^2}} \cdot dx$$

$$= [y]_0^1 \cdot 2 \cdot \left[\sin^{-1}\left(\frac{x}{2}\right)\right]_0^1.$$

$$= (1) \cdot 2\left[\sin^{-1}\left(\frac{1}{2}\right) - \sin^{-1}\left(\frac{0}{2}\right)\right].$$

$$= 2\left(\frac{\pi}{6}\right) \Rightarrow \frac{\pi}{3}.$$

Question # 12

Sol:

$x^2+y^2+z^2 = 4$ when $z=1$,

$x^2+y^2+1 = 4$

$x^2+y^2 = 4-1$

$x^2+y^2 = 3$.

so,

$D = \{(x,y) \mid x^2+y^2 \leq 3\}$,

and $z = \sqrt{4-x^2-y^2}$.

$\rightarrow f_x(x,y)$:

$= \frac{1}{\cancel{2}}(4-x^2-y^2)^{-1/2} \cdot (-\cancel{2}x)$

$= -x(4-x^2-y^2)^{-1/2}$.

$\rightarrow f_y(x,y)$:

$= \frac{1}{\cancel{2}}(4-x^2-y^2) \cdot (-\cancel{2}y)$

$= -y(4-x^2-y^2)$.

$\rightarrow$ Surface Area:

$$A(S) = \iint_D \sqrt{\left[(-x)(4-x^2-y^2)^{-1/2}\right]^2 + \left[(-y)(4-x^2-y^2)^{-1/2}\right]^2 + 1}\, dA$$

$$= \int_0^{2\pi}\int_0^{\sqrt{3}} \sqrt{\frac{x^2}{4-x^2-y^2} + \frac{y^2}{4-x^2-y^2} + 1} \cdot r\,dr \cdot d\theta$$

$$= \int_0^{2\pi}\int_0^{\sqrt{3}} \sqrt{\frac{r^2}{4-r^2} + 1} \cdot r \cdot dr \cdot d\theta.$$

$$= \int_0^{2\pi} \sqrt{\frac{\cancel{r^2} - 4 - \cancel{r^2}}{4-r^2}} \cdot r \cdot dr \cdot d\theta$$

$$= \int_0^{2\pi}\int_0^{\sqrt{3}} \frac{2r}{\sqrt{4-r^2}} \cdot dr \cdot d\theta$$

Let $\cancel{u = 4 - r^2}$

$$\cancel{du = \frac{1}{2}(4-r^2)^{-1/2} \cdot (-2r)}$$

$$\cancel{2du = (4-r^2)}$$

$$u = 4 - r^2$$

$$du = -2r \cdot dr$$

$$-du = 2r \cdot dr$$

$$= -\int_0^{2\pi}\int_0^{\sqrt{3}} \frac{1}{\sqrt{u}} \cdot du \cdot d\theta$$

$$= -\int_0^{2\pi}\int_0^{\sqrt{3}} (u)^{-1/2} \cdot du \cdot d\theta$$

$$\cancel{= \int_0^{2\pi} -2\left[u^{1/2}\right]_0^{\sqrt{3}}}$$

$$\cancel{= \int_0^{2\pi} -2\left[(4-r^2)^{1/2}\right]_0^{\sqrt{3}}}$$

$$\cancel{= \int_0^{2\pi} 2 \cdot \left[(4-(\sqrt{3})^2)^{1/2} - (4-0^2)^{1/2}\right]}$$

$$\cancel{= \int_0^{2\pi}}$$

$$= -\int_0^{2\pi} 2\left[u^{1/2}\right]_0^{\sqrt{3}} \cdot d\theta$$

$$= -\int_0^{2\pi} 2\left[\sqrt{4-r^2}\right]_0^{\sqrt{3}} \cdot d\theta$$

$$= -\int_0^{2\pi} 2\sqrt{4-(\sqrt{3})^2} - 2\sqrt{4-(0)^2} \cdot d\theta$$

$$= -\int_0^{2\pi} 2 - 4 \, d\theta$$

$$= \int_0^{2\pi} 2 \, d\theta$$

$$= 2\left[\theta\right]_0^{2\pi}$$

$$= 2(2\pi) \Rightarrow 4\pi.$$

Question 13

Sol:

$x^2 + y^2 + z^2 = a^2$ sphere Σ_0

cylinder $x^2 + y^2 = ax$. above xy plane.

$z = \sqrt{a^2 - x^2 - y^2}$.

→ partial derivative w.r.t x

$z_x = \frac{1}{2}(a^2 - x^2 - y^2)^{-1/2} \cdot (-2x)$

$= -x(a^2 - x^2 - y^2)^{-1/2}$

→ partial derivative w.r.t y:

$z_y = \frac{1}{2}(a^2 - x^2 - y^2)(-2y)$

$= (-y)(a^2 - x^2 - y^2)$.

→ Surface Area:

$$\iint_D \sqrt{\frac{x^2+y^2}{a^2-x^2-y^2} + 1}\, dA.$$

$$= \int_{\pi/2}^{-\pi/2} \int_{a}^{a\cos\theta} \sqrt{\frac{r^2}{a^2-r^2} + 1}\; r \cdot dr \cdot d\theta.$$

$$= \int_{\pi/2}^{\pi/2} \int_{0}^{a\cos\theta} \frac{a \cdot r}{\sqrt{a^2-r^2}} \cdot dr \cdot d\theta.$$

$$= \int_{-\pi/2}^{\pi/2} \left[-a \cdot \sqrt{a^2 - r^2}\right]_{r=0}^{r=a\cos\theta} \cdot d\theta.$$

$$= \int_{-\pi/2}^{\pi/2} -a\left(\sqrt{a^2 - a^2\cos^2\theta} - a\right) \cdot d\theta.$$

$$= 2a^2 \int_{0}^{\pi/2} \left(1 - \sqrt{1 - \cos^2\theta}\right) \cdot d\theta.$$

$$= 2a^2 \int_0^{\pi/2} d\theta - 2a^2 \int_0^{\pi/2} \sqrt{\sin^2\theta} \cdot d\theta$$

$$= a^2\pi - 2a^2 \int_0^{\pi/2} \sin\theta \cdot d\theta .$$

$$= a^2(\pi - 2)$$

Exercise 15.6

Question 3:

Sol:

$$= \int_0^2 \int_0^{z^2} \int_0^{y-z} (2x - y) \cdot dx \cdot dy \cdot dz$$

$$= \int_0^2 \int_0^{z^2} \left[x^2 - xy\right]_0^{y-z} \cdot dy \cdot dz$$

$$= \int_0^2 \int_0^{z^2} (y-z)^2 - (y-z)y \cdot dy \cdot dz$$

$$= \int_0^2 \int_0^{z^2} \cancel{y^2} + z^2 - 2yz - \cancel{y^2} + yz \cdot dy \cdot dz$$

$$= \int_0^2 \int_0^{z^2} (z^2 - yz) \cdot dy \cdot dz$$

$$= \int_0^1 \left[z^2 y - \frac{y^2 z}{2}\right]_0^{z^2} \cdot dz$$

$$= \int_0^1 z^2(z^2) - \frac{(z^2)^2(z)}{2} \cdot dz$$

$$= \int_0^1 z^4 - \frac{z^5}{2} \cdot dz$$

$$= \left[\frac{1}{5} z^5 - \frac{1}{12} \cdot z^6 \right]_0^2$$

$$= \frac{32}{5} - \frac{64}{12} \Rightarrow \frac{16}{15}$$

Question 6 :

$$\int_0^{\pi/2} \int_0^{2x} \int_0^{x+z} \cos(x - 2y + z) \cdot dy \cdot dz \cdot dx$$

$$= \int_0^{\pi/2} \int_0^{2x} -\frac{1}{2} \left[\sin(x - 2y + z) \right]_{y=0}^{y=x+z} \cdot dz \cdot dx$$

$$= -\frac{1}{2} \int_0^{\pi/2} \int_0^{2x} \left[\sin(x - 2(x+z) + z) - \sin(x+z) \right] \cdot dz \cdot dx$$

$$= -\frac{1}{2} \int_0^{\pi/2} \int_0^{2x} \sin(-x - z) - \sin(x+z) \cdot dz \cdot dx$$

$$= -\frac{1}{\cancel{2}} \int_0^{\pi/2} \int_0^{2x} -\cancel{2} \sin(x+z) \cdot dz \cdot dx$$

$$= \int_0^{\pi/2} \int_0^{2x} \sin(x+z) \cdot dz \cdot dx$$

$$= \int_0^{\pi/2} - \left[\cos(x+z) \right]_{z=0}^{z=2x}$$

$$\cos(x + 2x) - \cos(x) \cdot dx$$

$$= -\int_0^{\pi/2} \cos(x + 2x) - \cos(x) \cdot dx$$

$$= -\int_0^{\pi/2} \cos(3x) - \cos(x) \cdot dx$$

$$= \left[\sin x - \frac{1}{3}\sin 3x\right]_0^{\pi/2}$$

$$= 1 - \frac{1}{3}(-1)$$

$$= \frac{4}{3}$$

Question 18:

Sol:

$$= \int_0^1\int_0^1\int_0^{2-x^2-y^2} xye^z \cdot dz \cdot dy \cdot dx.$$

$$= \int_0^1\int_0^1 xy\left[e^z\right]_{z=0}^{z=2-x^2-y^2} \cdot dy \cdot dx$$

$$= \int_0^1\int_0^1 xy\left[e^{2-x^2-y^2} - e^0\right] \cdot dy \cdot dx$$

$$= \int_0^1\int_0^1 xy \cdot e^{2-x^2-y^2} - xy \cdot dy \cdot dx$$

$$= \int_0^1\left[\frac{-1}{2}xe^{2-x^2-y^2} - \frac{1}{2}xy^2\right]_{y=0}^{y=1} \cdot dx$$

$$= \int_0^1\left(\frac{-1}{2}xe^{2-x^2-1} - \frac{1}{2}x(1)^2 + \frac{1}{2}xe^{2-x^2}\right) \cdot dx.$$

$$= \int_0^1\left(\frac{-1}{2}xe^{1-x^2} - \frac{1}{2}x + \frac{1}{2}xe^{2-x^2}\right) dx$$

$$= \left[\frac{+1}{4}xe^{1-x^2} - \frac{1}{4}x^2 \div \frac{1}{4}x \cdot e^{2-x^2}\right]_0^1$$

$$= \frac{1}{4} - \frac{1}{4} - \frac{1}{4}e + 0 + \frac{1}{4}e^2$$

$$= \frac{1}{4}e^2 - \frac{1}{2}e.$$

Question 5:

Sol:

$$= \int_1^2 \int_0^{2z} \int_0^{\ln x} x \cdot e^{-y} \cdot dy\, dx \cdot dz.$$

$$= \int_1^2 \int_0^{2z} \left. -x e^{-y} \right]_{y=0}^{y=\ln x} \cdot dx \cdot dz$$

$$= \int_1^2 \int_0^{2z} \left(-x e^{-\ln(x)} + x e^{0}\right) \cdot dx \cdot dz$$

$$= \int_1^2 \int_0^{2z} (-1 + x) \cdot dx \cdot dz$$

$$= \int_1^2 \left[-x + \frac{1}{2} x^2\right]_{x=0}^{x=2z} \cdot dz$$

$$= \int_1^2 \left[-(2z) + \frac{1}{2}(2z)^2\right] - \left[0 + \frac{1}{2}(0)^2\right] \cdot dz$$

$$= \int_1^2 \left(-2z + 2z^2\right) \cdot d\cancel{x}.$$

$$= \left[-z^2 + \frac{2}{3} z^3\right]_1^2.$$

$$= \left(-(2)^2 + \frac{2}{3}(2)^3\right) - \left(-1^2 + \frac{2}{3}(1)^2\right.$$

$$= -4 + \frac{16}{3} + 1 - \frac{2}{3}$$

$$= \frac{5}{3}.$$

Question 13

Sol:

$$\iiint_E y \cdot dV$$

$$= \int_0^3 \int_0^x \int_{x-y}^{x+y} y \cdot dz \cdot dy \cdot dx$$

$$= \int_0^3 \int_0^x \left[yz \right]_{z=x-y}^{z=x+y} \cdot dy \cdot dx$$

$$= \int_0^3 \int_0^x 2y^2 \cdot dx$$

$$= \int_0^3 \left[\frac{2}{3} y^3 \right]_{y=0}^{y=x} \cdot dx$$

$$= \int_0^3 \frac{2}{3} x^3 \cdot dx$$

$$= \frac{1}{6} \left[x^4 \right]_{x=0}^{x=3}$$

$$= \frac{81}{6}$$

$$= \frac{27}{2}$$